# [PHYS 6268] Nonlinear Dynamics: Problem Set 8

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**Strogatz Book: Problems 7.5.4, 7.6.3, 7.6.6, & 7.6.17**

## 1 Problem 7.5.4: Piecewise-Linear Nullclines

Consider the equation

$$\ddot{x} + \mu f(x)\dot{x} + x = 0 \qquad , \tag{1}$$

where the function $f(x)$ is defined in the piecewise form:

$$f(x) = \begin{cases} -1; & |x| < 1 \\ +1; & |x| \geq 1 \end{cases}$$

### 1.1 (a) Show that the system is equivalent to $\dot{x} = \mu(y - F(x))$, $\dot{y} = -x/\mu$, where $F$ is piecewise linear.

To do so, we begin by obtaining a change of variables for the general form of the coefficient function $f$. It is important to recognize that $f$ is piecewise constant, that is, a sum of Heaviside functions. However, $f$ divides the coordinate domain $x \in \mathbb{R}$ into three separate "regions":

$$f(x) = \begin{cases} +1; & x \leq -1 \\ -1; & -1 < x < 1 \\ +1; & x \geq 1 \quad . \end{cases}$$

Consider the rewritten form of equation (1), given by

$$\ddot{x} + \mu\dot{x}f(x) = -x \quad .$$

Notice that

$$(\ddot{x} + \mu\dot{x}f(x)) = \frac{\mathrm{d}}{\mathrm{d}t}(\dot{x} + \mu F(x)) \quad ,$$

where $F(x)$ is some real-valued function that we will determine below. Defining the new variable

$$w \equiv \dot{x} + \mu F(x)$$

allows us to write (using equation (1)) the following,

$$\dot{w} = -x \qquad .$$

Then, the definition of $w$ implies that

$$\dot{x} = w - \mu F(x)) \qquad .$$

By replacing $w \mapsto \mu y$, we recover the system equivalent to equation (1) that was requested, given by

$$\begin{aligned} \dot{x} &= \mu(y - F(x)) \\ \dot{y} &= -\frac{x}{\mu} \ . \end{aligned}$$

However, this recasting was predicated on the existence of the function $F(x)$. Hence, we must calculate $F(x)$ explicitly for the different piecewise values of $f$ in order for the above to be valid. We begin with

$$\frac{\mathrm{d}}{\mathrm{d}t}(\dot{x} + \mu F(x)) = \ddot{x} + \mu \dot{x} f(x) = -x \quad .$$

By distributing the differential operator, we have

$$\implies \frac{\mathrm{d}}{\mathrm{d}t} \mu F(x) = \mu \dot{x} f(x) \qquad ,$$

which is an expression through which we can determine $F$. Notice that

$$\frac{\mathrm{d}}{\mathrm{d}t} F(x) = \frac{\mathrm{d}}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} F(x) = \dot{x} \frac{\mathrm{d}}{\mathrm{d}x} F(x) \quad .$$

Then, the above becomes

$$\frac{\mathrm{d}}{\mathrm{d}x} F(x) = f(x) \qquad \implies \qquad F(x) = \int f(x)\,\mathrm{d}x \qquad .$$

Therefore, using the definition of $\dot{x}$, we can define $F$ as the antiderivative of $f$.

This allows us to calculate $F(x)$ for all cases (i.e., sub-functions) of the piecewise function $f$ individually via direct integration. This yields

$$F(x) = \int f(x)\,\mathrm{d}x = \begin{cases} x + c_1 & x \leq -1 \\ -x + c_2 & -1 < x < 1 \\ x + c_3 & x \geq 1 \quad . \end{cases}$$

Here, the constants $c_1, c_2, c_3 \in \mathbb{R}$ of integration can be determined by mandating that $F(x)$ be piecewise continuous. This yields the conditions

$$x + c_1|_{x=-1} = -x + c_2|_{x=-1}$$

and

$$-x + c_2|_{x=1} = x + c_3|_{x=1} \qquad ,$$

which can be combined to yield

$$c_2 = \frac{c_1 + c_3}{2} \qquad .$$

We also know that since $f(x) = -1$ for $|x| < 1$, the value $F(-1) = F(1) + 2$. This implies that

$$c_1 = c_3 + 4 \qquad .$$

By letting $c_2 = 0$ (centering $F$ along the vertical axis), we find $c_1 = 2$, and $c_3 = -2$. Hence, we have shown that the system described by equation (1) is equivalent to

$$\begin{aligned} \dot{x} &= \mu(y - F(x)) \\ \dot{y} &= -\frac{x}{\mu} \quad , \end{aligned}$$

where

$$F(x) = \begin{cases} x + 2; & x \leq -1 \\ -x; & -1 < x < 1 \\ x - 2; & x \geq 1 \quad . \end{cases}$$

### 1.2 (b) Graph the nullclines.

The nullclines and associated phase portrait for the system are given in Figure 1. In order to generate the phase portrait, we let the parameter $\mu = 1$.

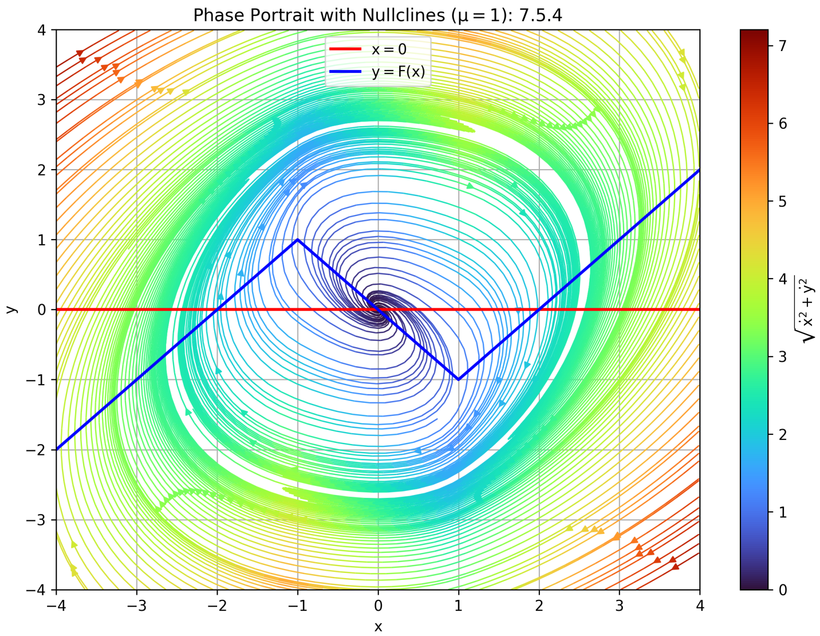


Figure 1: Nullclines and phase portrait for the system $\ddot{x} + \mu f(x)\dot{x} + x = 0$, recast in a change of variables. The nullcline line corresponding to $\dot{x} = 0$ is given in blue, and the line corresponding to $\dot{y} = 0 = -x$ is in red (see also Figure caption). The magnitude of the velocity field $(\dot{x}, \dot{y})$ corresponds to the colormap on the right. For the purposes of generating the Figure, the parameter $\mu$ is set to unity.

### 1.3 (c) Show that the system exhibits relaxation oscillations for $\mu \gg 1$, and plot the limit cycle in the $(x, y)$ plane.

To demonstrate the behavior of the system for $\mu \gg 1$, consider an arbitrary initial condition $(x_0, y_0)$ near the origin, that is, satisfying $y_0 > 0$ and $x_0^2 + y_0^2 \sim \mathcal{O}(1)$. Then, $y_0 - F(x_0) \sim \mathcal{O}(1)$ as well. Therefore, the gradient of the system at this initial condition follows the approximate behavior of

$$\begin{aligned}\dot{x} &\sim \mu\mathcal{O}(1) \sim \mathcal{O}(\mu)\\ \dot{y} &\sim -\mathcal{O}(1)/\mu \sim -\mathcal{O}(\mu^{-1})\,.\end{aligned}$$

Hence, the flow from this initial condition has a gradient directed almost entirely in the $x$ direction, since $\mathcal{O}(\mu) \gg 1$. The small drop in $y$ value is largely overshadowed by the left-to-right translation in the phase plane. This behavior is exhibited until the trajectory approaches the nullcline satisfying $y \approx F(x)$: at this point, the gradient rapidly changes order, following now

$$\begin{aligned}\dot{x} &\approx 0\\ \dot{y} &\sim -\mathcal{O}(x)/\mu\,.\end{aligned}$$

This largely vertical gradient steeply "turns" the system trajectories from moving largely in positive $x$ to largely in negative $y$. As soon as $y$ drops, however, the value of $\dot{x} = \mu(y - F(x))$ grows negative and the system tends diagonally down and to the left (towards $-x, -y$). As the trajectory approaches the "corner" of the nullcline at $(x, y) = (1, -1)$, the system again jumps to rapidly move towards negative values. As it reaches and crosses the nullcline, the cycle repeats. This cyclic trajectory with two separate timescales for the two regimes of the orbit is consistent with a relaxation oscillation.

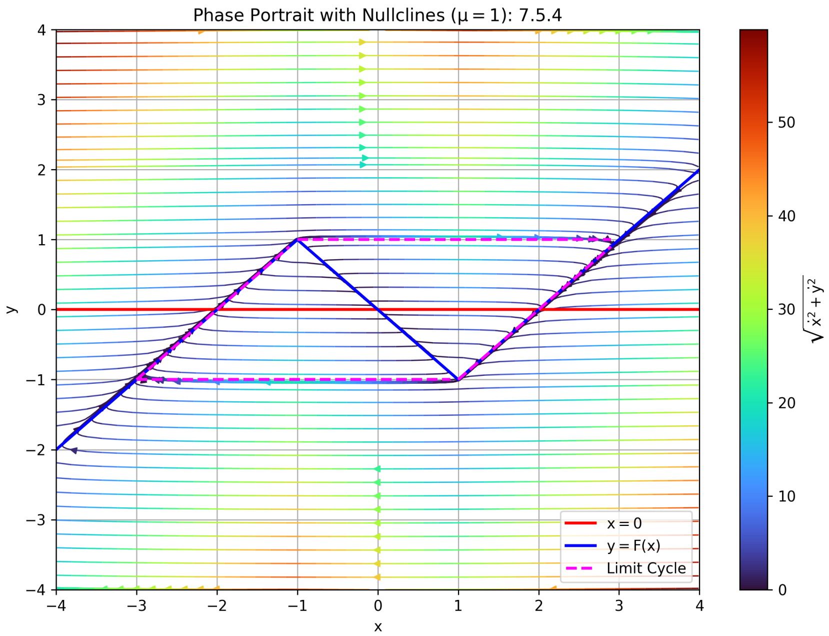


Figure 2: Limit cycle (magenta) for the system $\ddot{x} + \mu f(x)\dot{x} + x = 0$, in the limit of $\mu \gg 1$. To generate the Figure, we let $\mu = 10$. The nullcline line corresponding to $\dot{x} = 0$ is given in blue, and the line corresponding to $\dot{y} = 0 = -x$ is in red (see also Figure caption). The magnitude of the velocity field $(\dot{x}, \dot{y})$ corresponds to the colormap on the right.

To verify that this estimation for the system behavior is consistent with the analytical results, we plot the phase portrait for $\mu = 10 \gg 1$. This is shown in Figure 2, with the limit cycle orbit highlighted in magenta (dashed). As indicated by the difference between Figures 1 and 2, the system dynamics change drastically in the limit of $\mu \gg 1$.

### 1.4 (d) Estimate the period of the limit cycle for $\mu \gg 1$.

Since the timescale for the trajectory to travel along the nullclines in the orbit greatly exceed those of the fast "jumps" between branches, we can estimate the period of the limit cycle as roughly double the time taken for the trajectory to travel along a branch of the nullcline $y = F(x)$. To obtain an approximate value for this timescale, we note that $y \approx F(x)$. This allows us to obtain an approximate expression for $\mathrm{d}t$ in terms of coordinates that can be integrated. We have that

$$\dot{y} \approx \frac{\mathrm{d}}{\mathrm{d}t} F(x) = \dot{x} f(x) \qquad .$$

However, we know that

$$\dot{y} = -\frac{x}{\mu}$$

so using $\mathrm{d}y \approx \mathrm{d}x f(x)$ we find that

$$\mathrm{d}t = -\frac{\mu}{x}\mathrm{d}y = -\mathrm{d}x f(x)\frac{\mu}{x} \qquad .$$

The branch of the limit cycle we are interested in occurs in the range $x \in [1, 3]$. Then, the period of the limit cycle is approximately

$$T \approx 2 \int \mathrm{d}t = -\mu \int_3^1 \mathrm{d}x \, \frac{f(x)}{x} = -\mu \int_3^1 \frac{\mathrm{d}x}{x} \qquad .$$

Then, we integrate to obtain

$$T \approx 2\mu \ln(3) \qquad .$$

Strogatz emphasizes that this estimate can be improved (with nontrivial effort) with terms proportional to roots of the Airy function.

## 2 Problem 7.6.3: Calibrating Regular Perturbation Theory; More Calibration

Consider the initial value problem

$$\ddot{x} + x = \epsilon$$
$$x(0) = 1; \quad \dot{x}(0) = 0$$

### 2.1 (a) Solve the problem exactly

In order to solve this inhomogeneous second order differential equation with the given intial data, we first note that it can be re-cast as a coupled set of first order equations. If we let $\mathbf{x}$ be a column vector containing $x, y$ and $y \equiv \dot{x}$, our equations take the form

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ y \end{pmatrix} = \dot{\mathbf{x}} = \mathbb{A} \cdot \mathbf{x} + \begin{pmatrix} 0 \\ \epsilon \end{pmatrix} \quad ; \quad \mathbb{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad .$$

First, we investigate the homogeneous system $\dot{\mathbf{x}}_h = \mathbb{A} \cdot \mathbf{x}_h$. it follows that the matrix describing the homogeneous system has a trace $\mathrm{Tr}\,\mathbb{A} = 0$ and determinant $\det \mathbb{A} = 1$. Since the determinant is positive, this implies that the fixed point must be a spiral or a center. This system is also traceless, so the eigenvalues are complex conjugates and the origin is a neutral center. The eigenvalues of the matrix can be calculated as follows,

$$\lambda_{\pm} = \frac{\mathrm{Tr}\,\mathbb{A} \pm \sqrt{\mathrm{Tr}\,\mathbb{A}^2 - 4\det \mathbb{A}}}{2} = \pm i \qquad , \tag{2}$$

Confirming that we have an entirely oscillatory solution, with no viscosity (damping) of the system energy. This agrees with our knowledge from encountering this expression time and time again representing a simple harmonic oscillator. The lack of a $\dot{x}$ term is consistent with the lack of viscous damping implied by $\mathbb{A}$.

Then, the general solution to our equation is given by

$$x_h(t) = c_1 \sin t + c_2 \cos t \qquad ,$$

which can be verified to satisfy $\ddot{x} = -x$ for constants $c_1, c_2 \in \mathbb{R}$ to be determined by the initial data. However, before doing so, we must consider the inhomogeneous solution $x_i$:

$$\ddot{x}_i + x_i = \epsilon \qquad .$$

We first try the simplest solution, $x_i = \epsilon$. This immediately satisfies the above, so if it can be reconciled with the initial data, then it represents the inhomogeneous solution. Then,

$$x(t) = x_i(t) + x_h(t) = c_1 \sin t + c_2 \cos t + \epsilon \qquad .$$

Hence, we seek to determine $c_1, c_2$. We know that

$$x(0) = 1 = \epsilon + c_2 \qquad ,$$

and that

$$\dot{x}(0) = c_1 = 0 \qquad .$$

This allows us to write the complete solution as follows,

$$x(t) = (1-\epsilon)\cos t + \epsilon \qquad . \tag{3}$$

### 2.2 (b) Using regular perturbation theory, find $x_0$, $x_1$, and $x_2$ in the series expansion

The approach of regular perturbation theory is to assume the solution can be expanded in successive powers of $\epsilon \ll 1$, with only the low-order terms contributing the majority of the system dynamics. We let

$$x(t,\epsilon) = x_0(t) + \epsilon x_1(t) + \epsilon^2 x_2(t) + \mathcal{O}(\epsilon^3) \qquad ,$$

where $x_0, x_1, x_2$ are smooth functions of $t$. Neglecting terms $\mathcal{O}(\epsilon^3)$ and smaller, we obtain via substitution

$$(\ddot{x}_0 + x_0) + \epsilon(\ddot{x}_1 + x_1) + \epsilon^2(\ddot{x}_2 + x_2) = \epsilon \qquad .$$

This provides us a system of equations using each power of $\epsilon$, since each of the coefficients must independently amount to the coefficient of the governing equation for that power of $\epsilon$. Then, we obtain

$$\begin{aligned} \ddot{x}_0 + x_0 &= 0 \\ \ddot{x}_1 + x_1 &= 1 \\ \ddot{x}_2 + x_2 &= 0 \end{aligned}$$

which correspond to three decoupled harmonic oscillators, one with forcing and the other two a simple harmonic oscillator. Each order $x_0, x_1, x_2$ solution corresponds to equation (2.1) with different values for the coefficients: $c_{1,0}, c_{2,0}$ for $x_0$, $c_{1,1}, c_{2,1}$ for $x_1$, and $c_{1,2}, c_{2,2}$ for $x_2$. To determine these constants, we rely on the initial data and the substituted solution. The initial data yields the following conditions

$$\begin{aligned} c_{1,0} = c_{1,1} = c_{1,2} &= 0 \\ c_{2,0} + \epsilon(1 + c_{2,1}) + \epsilon^2 c_{2,2} &= 1 \ . \end{aligned}$$

The first condition eliminates the $\sin t$ term for each order solution. In analogy to the derivation of the equations for each order of $\epsilon$, the second condition mandates three independent conditions:

$$\begin{aligned} c_{2,0} &= 1 \\ 1 + c_{2,1} &= 0 \\ c_{2,2} &= 0 \qquad , \end{aligned}$$

which determine the individual solutions for each order:

$$\begin{aligned} x_0(t) &= \cos t \\ x_1(t) &= 1 - \cos t \\ x_2(t) &= 0 \ , \end{aligned}$$

which combine to yield the solution to the system via regular perturbation theory:

$$x(t,\epsilon) = \cos t + \epsilon(1 - \cos t)$$

equivalent to the general solution obtained with exact analytical methods.

## 2.3 (c) Explain why the perturbation solution does or doesn't contain secular terms

The solution obtained via regular perturbation theory does not contain secular terms. Secular terms arise from resonant forcing, that is, when a secular term arises in an individual equation of the series expansion solution. In order to construct a perturbation series solution, the expansion for equations representative of the harmonic oscillator is carried out for each order of the small parameter $\epsilon$. If the term including $\epsilon$ is a damped one, then it will evoke a secondary timescale corresponding to the relaxation of the solution in the viscous medium, on top of the timescale for oscillations. This long-term behavior stemming from the viscous relaxation term results in resonant forcing for the order 1 equation. In our case, however, the inhomogeneous term represents a constant "forcing", but *not* relaxation proportional to the state of the system at any given time. No resonant forcing terms arise in the equations of successive orders in the perturbative expansion, and thus, no secular behavior is assigned to the solution via this procedure. Therefore, the $\epsilon$ term only imparts a slight "shift" to the well-defined $2\pi-$periodic behavior of solutions to the simple harmonic oscillator (i.e., the homogeneous solutions of the system).

# 3 Problem 7.6.6: $\ddot{x} + x + \epsilon h(x, \dot{x}) = 0$

**For the case of $h(x,\dot{x}) = x\dot{x}$ and for $0 < \epsilon \ll 1$, calculate the averaged equations and analyze the long-term behavior of the system. Find the amplitude and frequency of any limit cycles for the original system. If possible, solve the averaged equations explicitly for $x(t,\epsilon)$, given initial conditions $x(0) = a$, $\dot{x}(0) = 0$.**

By considering the "two-timing" approach and decomposing a system using two *distinct* temporal coordinates, one can write the zeroth order solution in terms of functions $r, \phi$, the slowly varying amplitude and phase, respectively, as follows:

$$x(t,\epsilon) = x_0(t) + \mathcal{O}(\epsilon) \approx x_0(t) \qquad ; \qquad x_0(t) = r(T)\cos(\tau + \phi(T)) \quad .$$

Here, $\tau$ and $T$ are the fast and slow time coordinates, respectively. Then, using Fourier analysis to express $h$ in terms of trigonometric functions (or complex exponentials) and mandating that secular terms vanish, we obtain the *averaged* equations for $r$ and $\phi$ (see also equation (53) in Strogatz):

$$\begin{aligned} r' &= \frac{1}{2\pi}\int_0^{2\pi} h(\theta)\sin\theta\,\mathrm{d}\theta \equiv \langle h\sin\theta\rangle \\ r\phi' &= \frac{1}{2\pi}\int_0^{2\pi} h(\theta)\cos\theta\,\mathrm{d}\theta \equiv \langle h\cos\theta\rangle \end{aligned}$$

where $\theta = \tau + \phi$ and $h(\theta)$ is given by

$$x\dot{x} = h(x,\dot{x}) = h(r\cos(\tau+\phi), -r\sin(\tau+\phi)) = h(r\cos\theta, -r\sin\theta) = -r^2\sin\theta\cos\theta \quad .$$

Then, the averaged equations can be calculated directly using this result. This yields

$$r' = -\frac{r^2}{2\pi}\int_0^{2\pi}\sin^2\theta\cos\theta\,\mathrm{d}\theta = -r^2\langle\sin^2\theta\cos\theta\rangle \qquad .$$

This can be calculated via the following

$$\begin{aligned} r' &= -r^2\langle\sin^2\theta\cos\theta\rangle \\ &= -r^2\langle(1-\cos^2\theta)\cos\theta\rangle \\ &= -r^2\left(\langle\cos\theta\rangle - \langle\cos^3\theta\rangle\right) \\ &= r^2(0-0) \\ &= 0 \qquad . \end{aligned}$$

In a similar way, the other equation becomes

$$r\phi' = -\frac{r^2}{2\pi}\int_0^{2\pi} \sin\theta\cos^2\theta \,\mathrm{d}\theta = 0 \qquad .$$

Then, we have arrived at the system

$$\begin{aligned} r' &= 0 \\ \phi' &= 0\ . \end{aligned}$$

The system described by the above admits a constant solution, where neither the amplitude $r$ nor the phase $\phi$ of the solution $x(t)$ are sensitive to changes on the large timsecale $T$. Then, the solution becomes

$$x(t) \approx r(T)\cos(\tau + \phi(T)) = r_0\cos(\tau + \phi_0)$$

which reduces to the general solution of the simple harmonic oscillator with an arbitrary amplitude and phase shift set by the initial data. Using our initial conditions, it is implied $\phi_0 = 0$ and $r_0 = a$. This yields

$$x(t) \approx a\cos t$$

Hence, the solutions represent neutrally-stable orbits around the center with amplitude $a$. Since there is no relaxation or growth of orbits in amplitude, there is no limit cycle for this system. While there are periodic trajectories in phase space, there are no “nearby” trajectories to the solution that are attracted to the orbit as $t \longrightarrow \infty$. The period of the stable oscillations is $2\pi$, since the long term variability in phase $\phi$ vanishes for this case of $h(x,\dot{x}) = x\dot{x}$.

## 4 Problem 7.6.17: Playing on a Swing

A simple model for a child playing on a swing is

$$\ddot{x} + (1 + \epsilon\gamma + \epsilon\cos 2t)\sin x = 0 \qquad ,$$

where $\epsilon, \gamma \in \mathbb{R}$ and $\epsilon$ satisfies $0 < \epsilon \ll 1$. The variable $x$ measures the angle between the swing and the downward vertical. The term $(1 + \epsilon\gamma + \epsilon\cos 2t)$ models the effects of gravity and the periodic pumping of the child’s legs at approximately twice the natural frequency of the swing. The question is: starting near the fixed pint $x = 0$, $\dot{x} = 0$, can the child get the swing going by pumping their legs this way, or do they need a push?

### 4.1 (a)

**For small $x$, the equation may be replaced by**

$$\ddot{x} + (1 + \epsilon\gamma + \epsilon\cos 2t)x = 0 \qquad .$$

**Show that the averaged equations become**

$$\begin{aligned} r' &= \frac{1}{4}r\sin 2\phi \\ \phi' &= \frac{1}{2}\left(\gamma + \frac{1}{2}\cos 2\phi\right) \qquad , \end{aligned}$$

where $x = r\cos\theta = r(T)\cos(t + \phi(T))$, $\dot{x} = -r\sin\theta = -r(T)\sin(t + \phi(T))$, and instances of “prime” denote differentiation with respect to slow time $T = \epsilon t$.

We begin by reconfiguring our system into a form akin to that in problem 3. The above equation is equivalent to

$$\ddot{x} + x + \epsilon(\gamma + \cos 2t)x = 0 \qquad .$$

Hence, in the above formalism and in that of Strogatz, defining the function

$$h(x, \dot{x}) = (\gamma + \cos 2t)x$$

allows us to write

$$\ddot{x} + x + \epsilon h(x, \dot{x}) = 0 \qquad .$$

Then, to compute the averaged equations, we must write $h$ in terms of $\theta$:

$$\begin{aligned} h(x, \dot{x}) &= (\gamma + \cos 2t)x \\ \implies h(\theta) &= r\cos\theta(\gamma + \cos 2t) \\ &= r\cos\theta(\gamma + \cos 2(\theta - \phi)) \\ &= \gamma r\cos\theta + r\cos\theta(\cos 2\theta\cos 2\phi + \sin 2\theta \sin 2\phi)\ . \end{aligned}$$

Now, we can compute the averaged equations from their definition. This yields

$$r' = \gamma r\langle\sin\theta\cos\theta\rangle + r\langle\sin\theta\cos\theta(\cos 2\theta\cos 2\phi + \sin 2\theta\sin 2\phi)\rangle$$

which is equal to

$$r' = \frac{r}{2}\langle\sin 2\theta(\cos 2\theta\cos 2\phi + \sin 2\theta\sin 2\phi)\rangle \quad .$$

When expanded, this reads

$$r' = \frac{r}{2}\langle\sin 2\theta\cos 2\theta\cos 2\phi\rangle + \frac{r}{2}\langle\sin^2 2\theta\sin 2\phi\rangle \quad ,$$

which implies

$$r' = \frac{r}{4}\sin 2\phi \quad .$$

Similarly, for $\phi'$ we write:

$$r\phi' = \gamma r\langle\cos^2\theta\rangle + r\langle\cos^2\theta(\cos 2\theta\cos 2\phi + \sin 2\theta\sin 2\phi)\rangle$$

which simplifies to

$$r\phi' = \frac{\gamma r}{2} + r\langle\cos^2\theta\cos 2\theta\cos 2\phi\rangle \qquad ,$$

amounting to the result

$$\phi' = \frac{\gamma}{2} + \frac{\cos 2\phi}{4} \qquad .$$

Hence, the averaged equations are consistent.

### 4.2 (b) Show that the fixed point $r = 0$ is unstable to exponentially growing oscillations

That is, we must show that

$$r(T) = r_0 e^{kT} \qquad k > 0,$$

if $|\gamma| < \gamma_c$, where $\gamma_c$ is to be determined.

We first note that since $r' \propto r$, $\phi' \gg r'$ for trajectories near the origin $r = 0$. Thus, the system drives spiraling behavior more rapidly as trajectories approach the origin. This implies that $\phi$ equilibriates rapidly, or rather, that the evolution of the system in $\phi$ occurs “before” the behavior in $r$ changes much.

Hence, we may treat $\phi$ as a constant while $r$ continues to change. Then, we can solve the equation for $r$ as if it were not coupled to $\phi$, and use the value of $\phi$ corresponding to the equilibrium in the $\phi$ coordinate. This reads

$$r' = \frac{r}{4}\sin 2\phi \quad \Longrightarrow \quad \frac{\mathrm{d}r}{\mathrm{d}T} = \frac{r}{4}\sin 2\phi \quad .$$

Then, we can separate

$$\frac{\mathrm{d}r}{r} = \frac{\sin 2\phi}{4}\mathrm{d}T \quad ,$$

which can be directly integrated:

$$\Longrightarrow \ln r = \frac{\sin 2\phi}{4}T \quad .$$

This yields an exponential solution according to

$$r(T) = r_0 e^{\frac{\sin 2\phi}{4}T} \quad .$$

Now, we turn to the $\phi$ component, which equilibrates very rapidly so we consider to vanish after a short time:

$$\phi' = 0 = \frac{\gamma}{2} + \frac{\cos 2\phi}{4} \quad .$$

Hence, $\cos 2\phi = -2\gamma$. Using $\sin^2 n + \cos^2 n = 1$ implies a value for $\sin 2\phi$, corresponding to

$$\sin 2\phi = \sqrt{1 - 4\gamma^2} \quad . \tag{4}$$

Then, since the term in the exponential $k \equiv \frac{\sin 2\phi}{4} > 0$, the system exhibits exponentially growing oscillations stemming from the fixed point at the origin.

The critical value of the parameter $\gamma_c$ causes the system behavior to change at $\gamma = \pm\gamma_c$, that is, it corresponds to when equation (4) vanishes. Thus,

$$\gamma_c = \frac{1}{2} \quad .$$

### 4.3 (c) For $|\gamma| < \gamma_c$, determine the growth rate $k$ in terms of $\gamma$

From the above analysis, the growth rate is given by

$$k = \frac{\sin 2\phi}{4} = \frac{\sqrt{1 - 4\gamma^2}}{4}$$

### 4.4 (d) How do the solutions to the averaged equations behave if $|\gamma| > \gamma_c$?

In the case that $|\gamma| > \gamma_c$, the value of $\sin 2\phi$ becomes imaginary, assuming that the decoupled evolution of $\phi$ and $r$ remains. If the growth rate $k \in \mathbb{C}$ is complex, the value of $r$ will oscillate (rotating through the complex plane). Thus, trajectories will spiral with an oscillating radial amplitude.

### 4.5 (e) Interpret the results physically

The parameter $\gamma$ corresponds to the effect of the constant gravitational force acting downward to damp the swing oscillations. It represents the ratio of gravitational acceleration and the acceleration due to the child's legs pumping. The physical interpretation of the solution is as follows.

There are two cases. (i) When $|\gamma| < \gamma_c$, the system exhibits an exponentially growing amplitude. This corresponds to the swings of the child growing successively larger. However, this is still only valid in the small amplitude limit where $x \approx \sin x$, where we have carried out the above analysis. We also assume here that the phase separation $\phi$ approaches a constant value rapidly. On the other hand, in case (ii) the value of $|\gamma| > \gamma_c$. This corresponds to the configuration of the system where the child's pumping is being overcome by the gravitational force. The swing will oscillate around small amplitudes without any consistent growth, and the swinging will appear to a bystander less structured.

## 5 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the plotting: this was completed using python (`numpy` and `matplotlib`).